Confluent Cloud Terraform Deployment — User Event Enricher PTF UDF

> This example deploys the **User Event Enricher** PTF UDF (source in examples/ptf_udf_row_driven/java/) to **Confluent Cloud** using Terraform. All infrastructure (environment, Kafka cluster, Flink compute pool, service accounts, API keys) and Flink SQL statements are declared as Terraform resources.

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1.0 Overview

1.1 How this differs from the other deployment paths

Aspect	Confluent Cloud Terraform (this example)	Confluent Platform SQL Client
Where it runs	Confluent Cloud	Confluent Platform + Minikube
How SQL is submitted	`confluent_flink_statement` Terraform resources	`sql-client.sh -f` on the JobManager pod
UDF JAR delivery	`confluent_flink_artifact` (uploaded to CC)	`kubectl exec` to Flink pods
Entry point	`make deploy-cc-ptf-udf-row-driven`	`make deploy-cp-ptf-udf-row-driven`
Requires code compilation	✅ (Java + Gradle for UDF JAR)	✅ (Java + Gradle for UDF JAR)
Statement lifecycle	Managed by Terraform state	Managed by Flink session cluster

Aspect	Confluent Cloud Terraform (this example)	Confluent Platform SQL Client
Same codebase for both?	✅ (**same Java UDF code**, different Terraform vs SQL Client for deployment)	✅ (**same Java UDF code**, different Terraform vs SQL Client for deployment)

2.0 How it works

2.1 Infrastructure provisioning

Terraform declares the full Confluent Cloud stack:

Resource	Purpose
`confluent_environment`	Isolated CC environment (`ptf-udf`)
`confluent_kafka_cluster`	Standard single-zone Kafka cluster (AWS us-east-1)
`confluent_schema_registry_cluster`	Stream Governance Essentials (auto-provisioned with environment)
`confluent_service_account`	`ptf-udf` service account with FlinkDeveloper, ResourceOwner, and Assigner roles
`confluent_flink_compute_pool`	Flink compute pool (max 10 CFU)
`confluent_api_key`	Flink-specific API key for statement submission
`flink_api_key_rotation` module	Rotating API key pair for the Flink service account

2.2 JAR delivery

On Confluent Cloud, UDF JARs are uploaded as **Flink artifacts** via the `confluent_flink_artifact` resource. The JAR is built from examples/ptf_udf_row_driven/java/ and uploaded directly from the local build output. The `CREATE FUNCTION ... USING JAR` statement references the artifact using a `confluent-artifact://` URI.

2.3 Statement flow

Terraform manages the SQL statements as `confluent_flink_statement` resources with explicit `depends_on` ordering:

```
┌──────────────────────────────────────────────────────────────────┐
│  Step 1:  DROP TABLE IF EXISTS user_events              → OK      │
│  Step 2:  CREATE TABLE user_events (...)                → OK      │
│  Step 3:  INSERT INTO user_events VALUES (sample data)  → submitted │
│  Step 4:  DROP TABLE IF EXISTS enriched_events          → OK      │
│  Step 5:  CREATE TABLE enriched_events (...)            → OK      │
│  Step 6:  Upload UDF JAR as confluent_flink_artifact    → OK      │
│  Step 7:  CREATE FUNCTION user_event_enricher           → OK      │
```

```
│            USING JAR 'confluent-artifact://...'                          │
│  Step 8:   INSERT INTO enriched_events                      → submitted  │
│            SELECT ... FROM TABLE(user_event_enricher())                  │
└──────────────────────────────────────────────────────────────────────────┘
```

Step 8 is a **long-running streaming job**. It runs continuously, reading from `user_events` and writing enriched output to `enriched_events`.

Once deployment completes, Terraform generates a visual **resource graph** at `examples/ptf_udf_row_driven/cc_deploy/terraform.png`, providing an at-a-glance view of the infrastructure and resource dependencies:

3.0 Prerequisites

- macOS with Homebrew
- Java 17+
- Terraform installed
- A Confluent Cloud account with a **Cloud API key** and **secret** (create one here)
- The UDF JAR must be built before deploying:

```
make build-ptf-udf-row-driven   # builds the UDF fat JAR from
examples/ptf_udf_row_driven/java/
```

4.0 How to run

All commands are run from the **project root** (where the `Makefile` lives).

4.1 Deploy

A single target builds the UDF JAR and runs `terraform apply`:

```
make deploy-cc-ptf-udf-row-driven CONFLUENT_API_KEY=<your-key>
CONFLUENT_API_SECRET=<your-secret>
```

Behind the scenes this runs:

Step	What it does
1	`./gradlew clean shadowJar` — builds the UDF fat JAR from `examples/ptf_udf_row_driven/java/`
2	`terraform init` — initializes the Terraform working directory

Step	What it does
3	`terraform apply -auto-approve` — provisions all CC infrastructure and submits Flink SQL statements
4	Generates a Terraform visualization at `docs/images/terraform-visualization.png`

4.2 Monitor

Monitor the running Flink statements in the Confluent Cloud Console:

1. Navigate to your **ptf-udf** environment
2. Open the **Flink** tab to see compute pools and running statements
3. Open the **Topics** tab to inspect `user_events` and `enriched_events`

4.3 Tear down

To destroy all Confluent Cloud resources created by Terraform:

```
make teardown-cc-ptf-udf-row-driven CONFLUENT_API_KEY=<your-key>
CONFLUENT_API_SECRET=<your-secret>
```

This runs `terraform destroy -auto-approve`, removing all Flink statements, the compute pool, Kafka cluster, service accounts, and the environment.

5.0 Resources

- Confluent Terraform Provider
- confluent_flink_statement Resource
- confluent_flink_artifact Resource
- Create a User-Defined Function with Confluent Cloud for Apache Flink